

3.1 Defining the Derivative Worksheet

- Find $f'(x)$ if $f(x) = x^3 - 5x$.
- Consider the function $f(x) = 4x^2 - 9$.
 - Find an equation that will tell you slope of any tangent line on the graph of $f(x) = 4x^2 - 9$.
 - Construct the equation of a line that is tangent to $f(x)$ at the point where $x = \frac{1}{4}$.

3. Find the derivative of $g(x) = \frac{1}{x+1}$.

4. Find an equation of the tangent line to the curve at the given point.

a) $f(x) = 4x - 3x^2$, $(2, -4)$

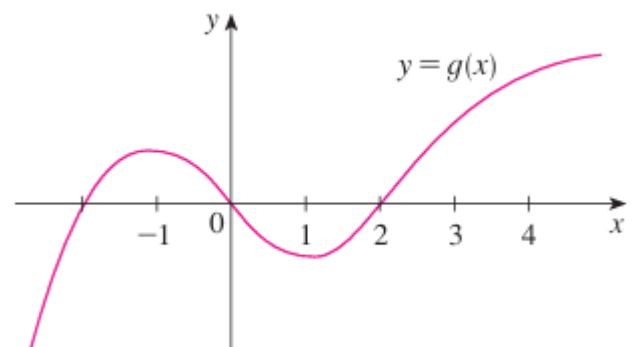
b) $f(x) = \sqrt{x}$, $(1, 1)$

c) $f(x) = x^3 - 3x + 1, (2, 3)$

d) $f(x) = \frac{2x+1}{x+2}, (1, 1)$

5. For the function $g(x)$ whose graph is shown below, arrange the following numbers in increasing order.

0 $g'(-2)$ $g'(0)$ $g'(2)$ $g'(4)$



6. The displacement (in meters) of a particle moving in a straight line is given by the equation $s = \frac{1}{t^2}$, where t is measured in seconds. Find the velocity of the particle at times:
- a) $t = 1$

b) $t = 2$

c) $t = 3$

d) $t = a$

7. The displacement (in meters) of a particle moving in a straight line is given by $s = t^2 - 8t + 18$, where t is measured in seconds.

a) Find the average velocity of the particle over each time interval:

i. $[3, 4]$

ii. $[3.5, 4]$

iii. $[4, 5]$

iv. $[4, 4.5]$

b) Find the instantaneous velocity when $t = 4$.

c) Sketch a picture of the function s and draw the secant lines whose slopes are the average velocities in part (a) and the tangent line whose slope is the instantaneous velocity in part (b).

8. A warm can of soda is placed in a cold refrigerator. Sketch the graph of the temperature of the soda as a function of time. Is the initial rate of change of the temperature greater or less than the rate of change after an hour?

9. The cost of producing x ounces of gold from a new gold mine is $C = f(x)$ dollars.

a) What is the meaning of the derivative $f'(x)$? What are its units?

b) What does the statement $f'(800) = 17$ mean?

10. The number of bacteria after t hours in a controlled laboratory experiment is $n = f(t)$.

a) What is the meaning of the derivative $f'(5)$? What are its units?

b) Suppose there is an unlimited amount of space and nutrients for the bacteria. Which do you think is larger, $f'(5)$ or $f'(10)$? If the supply of nutrients is limited, would that affect your conclusion? Explain.

11. Each of the following limits represents the derivative of some function f at some number a . State the function f and value of a in each case.

a) $\lim_{h \rightarrow 0} \frac{(1+h)^{10} - 1}{h}$

b) $\lim_{h \rightarrow 0} \frac{\sqrt[4]{16+h} - 2}{h}$

c) $\lim_{h \rightarrow \frac{\pi}{4}} \frac{\tan(x) - 1}{x - \frac{\pi}{4}}$

d) $\lim_{h \rightarrow 0} \frac{\cos(\pi + h) + 1}{h}$

e) $\lim_{t \rightarrow 1} \frac{t^4 + t - 2}{t - 1}$